CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Biometric face authentication has become increasingly prevalent in web-based identity verification, online examination supervision, financial onboarding, and secure login systems. Its convenience and contactless interaction make face recognition highly suitable for browser-integrated security workflows.

The landscape of digital security has undergone a significant evolution. For decades, the primary method of authentication was knowledge-based: passwords, PINs, and security questions. In response, the industry has shifted towards biometric authentication, which leverages unique physiological or behavioural characteristics of an individual for identity verification.

However, these systems are inherently vulnerable to presentation attacks (also known as spoofing), where adversaries attempt to gain unauthorized access by presenting printed photographs or digital screen replays to the camera. Such attacks are inexpensive to perform and require no technical skill, making spoof detection a critical component of any practical liveness verification system.

1.2 RELEVANCE

Facial recognition systems, while convenient, are highly susceptible to these presentation attacks. A simple, low-cost attack can be mounted by just printing a photo of the authorized user or displaying their picture on a smartphone screen. The success of such attacks can lead to severe consequences, including unauthorized access to sensitive personal data, financial accounts, and secure physical locations.

Therefore, it is not enough for a system to simply recognize a face; it must also verify that the face belongs to a live human being present at the point of capture. This crucial capability is known as face liveness detection or anti-spoofing.

Biometric authentication systems, particularly those relying on facial recognition, are increasingly deployed in government digital service portals and secure web-based login systems. Existing liveness detection solutions often rely on cloud-based verification platforms or computationally heavy deep learning models that must run on the server side, leading to privacy concerns, increased latency, and deployment limitations.

Therefore, there is a need for a lightweight, real-time, client-side spoof detection method capable of running directly inside a web browser, without depending on third-party verification services or high-performance computing hardware.

1.3 MOTIVATION

Government service portals and secure authentication systems are increasingly shifting toward web-based identity verification to improve accessibility and reduce manual verification overhead. However, integrating external liveness detection APIs introduces data privacy exposure, recurring subscription costs, and creates dependency on external service availability.

Meanwhile, existing deep learning-based spoof detection models—though accurate—are frequently too large or computationally demanding for real-time inference on the browser or low-power devices. This motivates the development of a bespoke liveness detection model that runs entirely in-browser using frameworks such as ONNX Runtime or TensorFlow.js, preserves user privacy by preventing image data from leaving the device, and offers reliable spoof detection accuracy across diverse real-world conditions (lighting variations, camera quality differences, device screen glare, and facial appearance diversity).

1.4 OBJECTIVES

The primary objectives of the proposed work are:

- To design a robust deep learning architecture (**SpoofNet**) specialized for detecting screen-based and printed spoof attacks.
- To develop a diverse custom dataset containing real and spoof samples captured using multiple device types and varied environmental conditions.
- To achieve high classification accuracy, ensuring minimal false acceptance of spoof attempts.
- To maintain fast inference speed suitable for real-time browser execution.

1.5 ORGANIZATION OF PROJECT REPORT

- **Chapter 1** introduces the fundamental concepts of biometric security, the problem of spoofing, and the motivation and objectives of this project.
- **Chapter 2** provides a survey of existing literature, a comparative analysis of common techniques, and a detailed review of key papers in the field.
- **Chapter 3** presents a detailed breakdown of the system's architecture, the proposed "SpoofNet" model, the custom dataset creation and splitting process, and the training methodology.
- **Chapter 4** details the implementation environment, software libraries, and provides key code snippets for the data preparation and training loop..
- **Chapter 5** concludes the report with a summary of the project's achievements and discusses promising directions for future enhancements.
- **Appendix A** provides the complete, runnable Python script for the final real-time inference application, which includes the SpoofNet model definition and all security checks.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The challenge of face liveness detection has been approached from various angles, with methods evolving in sophistication over time. Traditional liveness detection techniques have relied on hand-crafted texture descriptors, motion cues, or color distortion patterns. Traditional liveness detection techniques, relying on hand-crafted descriptors or motion cues, lack robustness to variations in illumination, camera quality, and spoof media type. Deep learning approaches demonstrate stronger generalisation, but widely adopted architectures, For example ResNet-50 and EfficientNet-B0 are not inherently optimised for spoof-specific cues [1][2].

Research in face anti-spoofing has evolved across two parallel axes: increasing detection accuracy and improving computational efficiency. Table 1 describes Early methods focused on hand-crafted texture descriptors, such as Local Binary Patterns (LBP), which provided a baseline for discriminating between live and spoofed textures [3]. However, these manual features lack robustness to diverse conditions. The advent of deep learning led to CNN-based solutions that automated feature extraction. To improve accuracy, many researchers explored multimodal data or complex auxiliary tasks.

For instance, two-stream CNN to incorporate depth estimation, while a CNN-RNN with auxiliary supervision from both depth and rPPG (remote photoplethysmography) signals [4] [5]. Similarly, FeatherNet achieved high performance by fusing RGB, depth, and IR data [6]. While effective, these methods introduce significant complexity, requiring specialised sensors (Depth/IR) or video-stream analysis (rPPG), making them unsuitable for deployment on standard, RGB-only webcams. Concurrently, a separate field of research focused on creating general-purpose, lightweight CNN architectures for efficient on-device execution. Models like SqueezeNet and MobileNetV2 achieved dramatic reductions in parameters and computational cost [7][8]. While these models are highly efficient, they are designed as general-purpose classifiers, not specifically architected to detect the subtle, high-frequency artefacts.

Our work, SpoofNet, addresses these critical deployment gaps:

- **SpoofNet vs. FeatherNets:** Achieves high accuracy without the need for specialized hardware like depth or IR sensors.
- **SpoofNet vs. Auxiliary Methods:** Relies on a single binary classification task, avoiding the complexity of multi-task training (e.g., depth estimation, rPPG) or video sequence requirements.
- **SpoofNet vs. Traditional LBP:** Uses automatic feature learning, which is more robust and adaptable than manual feature engineering.

The result is a production-ready anti-spoofing solution designed for standard RGB cameras, balancing accuracy and deployment efficiency.

2.2 LITERATURE SURVEY

Table 2.1: Literature Review of Early Implementation

Paper No	Architecture	Hardware	Training Approach	Key Limitation	Outcome
[3]	Handcrafted LBP texture features + SVM classifier	Standard RGB images	Manual feature engineering	Limited adaptability without manual tuning	~15% HTER on REPLAY-ATTACK;
[4]	Two-stream CNN: patch analysis + depth estimation	Standard RGB images	Synthetic depth map as auxiliary task	Dual CNN streams increase complexity	Pioneered depth auxiliary supervision
[5]	CNN-RNN with auxiliary supervision (depth + rPPG)	Standard RGB video	Multi-task learning with pseudo depth and rPPG labels	Requires video input; complex multi-task training	Improved cross-database generalizatio n
[6]	Lightweight CNN (0.35M parameters)	RGB + Depth + IR multi-modal sensors	Trained on multi-modal data	Requires depth/IR sensors for best performance	Highly efficient; ACER 0.00168
Proposed Project work(Spoof Net)	5-block progressive CNN (32→64→128→256→512) + batch norm + dropout	Standard RGB images only (224×224)	Single binary classification	4.88M parameters, 19MB model	99.39% validation accuracy

2.3 DETAIL ANALYSIS OF PRIOR WORK

To better contextualize our proposed method, a more detailed examination of the key papers from the comparative analysis is presented below.

LBP for Anti-spoofing is a foundational study in texture-based anti-spoofing. The core idea is that spoof artifacts (like print media or digital screens) have different micro-texture patterns than live human skin.

The authors used Local Binary Patterns (LBP), a powerful texture descriptor, to extract a feature vector from the face image. This feature vector was then fed into a Support Vector Machine (SVM) classifier to distinguish between "real" and "spoof". While effective for certain attack types and a significant improvement at the time, this method's reliance on a single, handcrafted feature makes it brittle.

Its performance degrades significantly with variations in lighting, camera focus, and the quality of the spoof medium.

Patch and Depth-based CNNs introduced the idea of using auxiliary information to help the network. Instead

of just classifying the whole image, the system used a two-stream CNN. One stream analyzed local patches of the face, looking for fine-grained spoof artifacts. The second, more novel stream was trained to estimate a "synthetic" depth map from the 2D image.

The hypothesis was that a live face is 3D, while a photo or screen is 2D (planar). By forcing the network to learn this auxiliary task of depth estimation, it became better at distinguishing 2D spoofs. The limitation was the complexity of the dual-stream architecture and the unreliability of "synthetic" depth.

Auxiliary Supervision (Depth + rPPG) work took the auxiliary task concept even further. It proposed a complex CNN-RNN (Recurrent Neural Network) architecture that processed video frames instead of static images. The network was trained to perform three tasks simultaneously:

1. **Binary Classification:** The main task (real or spoof).
2. **Depth Estimation:** Similar to Atoum et al., to check for 3D shape.
3. **rPPG Signal Prediction:** This was the most novel part. Remote Photoplethysmography (rPPG) is a technique to detect the human pulse (blood flow) from subtle color changes in the skin, which are invisible to the naked eye but detectable by a camera. A spoof attack would not have a live pulse. By using video (RNN) and these powerful auxiliary cues, the system achieved excellent cross-database generalization. However, its primary drawbacks are its high complexity and the mandatory requirement for video input, making it less suitable for quick, on-device static image verification.

FeatherNets focused on the deployment challenge, similar to our motivation. The authors argued that most models were too large and slow for on-device use. They proposed "FeatherNets," an extremely lightweight CNN architecture with only 0.35 million parameters. To achieve high accuracy with such a small model, they relied on multi-modal data, fusing information from RGB (color), Depth, and Infrared (IR) sensors. The fusion of these data streams made the anti-spoofing task much easier.

Summary of Gaps: The literature shows a clear trade-off:

- Traditional methods (LBP) are lightweight but not robust.
- Complex deep learning methods (Liu et al.) are accurate but too heavy and require video.
- Multi-modal methods (FeatherNets) are efficient but require specialized hardware.

Our proposed **SpoofNet** aims to fill the gap for a solution that is **(1)** accurate, **(2)** lightweight for on-device deployment, and **(3)** works with only a standard RGB camera.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

3.1 INTRODUCTION

The proposed face anti-spoofing system adopts a multi-stage pipeline designed for browser-based deployment, emphasizing both accuracy and computational efficiency. The complete system architecture comprises four primary stages:

1. **Stage 1: Face Detection and Localization:** The system employs **MediaPipe Face Mesh** for real-time facial landmark detection, which identifies 478 3D facial landmarks with high precision. This lightweight framework, part of the MediaPipe SDK, is specifically chosen for its high performance (low latency) and compatibility with browser environments (JavaScript) and mobile devices. Its ability to provide a dense 3D mesh, rather than just a simple bounding box, is crucial for accurate ROI extraction.

2. **Stage 2: Region of Interest (ROI) Extraction:** Using the detected facial landmarks, the system extracts a precise facial region. This is not a simple crop. It spans from the forehead (landmark 10) to the chin (landmark 152), with lateral boundaries defined by the left and right cheek edges (landmarks 234 and 454). A margin of 12% is added horizontally and 3% vertically from the top to ensure complete facial coverage while minimizing background interference. This precise cropping ensures the model's attention is focused on facial texture, not background noise.

3. **Stage 3: Deep Learning Classification:** The cropped and preprocessed facial region is fed into **SpoofNet**, our custom-designed convolutional neural network. The model is optimized for detecting texture-based artifacts from printed photos (e.g., paper texture, halftone patterns, lack of depth) and digital screens (e.g., moiré patterns, pixel grids, light reflection). The model outputs a two-class probability distribution (real vs. spoof).

4. **Stage 4: Post-Processing and Security Verification:** A raw model output is not robust enough for a security application. To enhance robustness against adversarial conditions and transient artifacts, the system implements multiple security mechanisms:

    - **Confidence Thresholding:** A strict threshold of 80% (0.8) is enforced for the "real" class. Any prediction below this is treated as uncertain or spoof.
    - **Temporal Consistency Verification:** A sliding window buffer of 10 consecutive frames is maintained. For a final "Pass" decision, at least 80% of the frames in this buffer (i.e., 8 out of 10) must agree. This effectively filters out false positives or negatives caused by momentary lighting changes, motion blur, or model instability.
    - **Quality Assurance Checks:** Before even sending a frame to the model, a series of checks are performed (detailed in Table 3.1) to reject poor-quality inputs.

3.2 MODEL ARCHITECTURE (SPOOFNET)

SpoofNet is a custom convolutional neural network specifically designed for facial anti-spoofing. It is not a general-purpose model like ResNet; it is deliberately designed with a lean architecture to be efficient for browser-based deployment while being deep enough to capture the subtle texture-based artifacts inherent in printed photos and digital screens.

The final, runnable code for this model is included in the inference script in Appendix A.

3.2.1 Network Design Philosophy

The architecture philosophy is driven by three core principles:

1. **Hierarchical Feature Extraction:** The model uses a progressive increase in feature channels (32 → 64 → 128 → 256 → 512) across five blocks. This allows the network to capture low-level, fine-grained texture details (e.g., screen moiré patterns, print artifacts, skin pores) in the early layers, while learning high-level semantic representations (e.g., facial liveness, depth, surface properties) in deeper layers.

2. **Aggressive Regularization:** To prevent overfitting on the 38,000-image dataset, the model employs extensive regularization. Batch Normalization is used after every convolutional and linear layer to stabilize training. Spatial Dropout (Dropout2d) is used in convolutional blocks, and Standard Dropout is used in the classifier. This forces the model to learn redundant features and improves its generalization to unseen spoofing conditions.

3. **Compact Representation:** The key to a small model size is the use of Global Average Pooling (GAP). After the final convolutional block, the 7x7x512 feature map is collapsed into a single 512-dimensional vector by averaging. This replaces a potentially massive Fully Connected layer, drastically reduces the parameter count, and makes the model more robust to minor translations in the input face.

3.2.2 ARCHITECTURE COMPONENTS

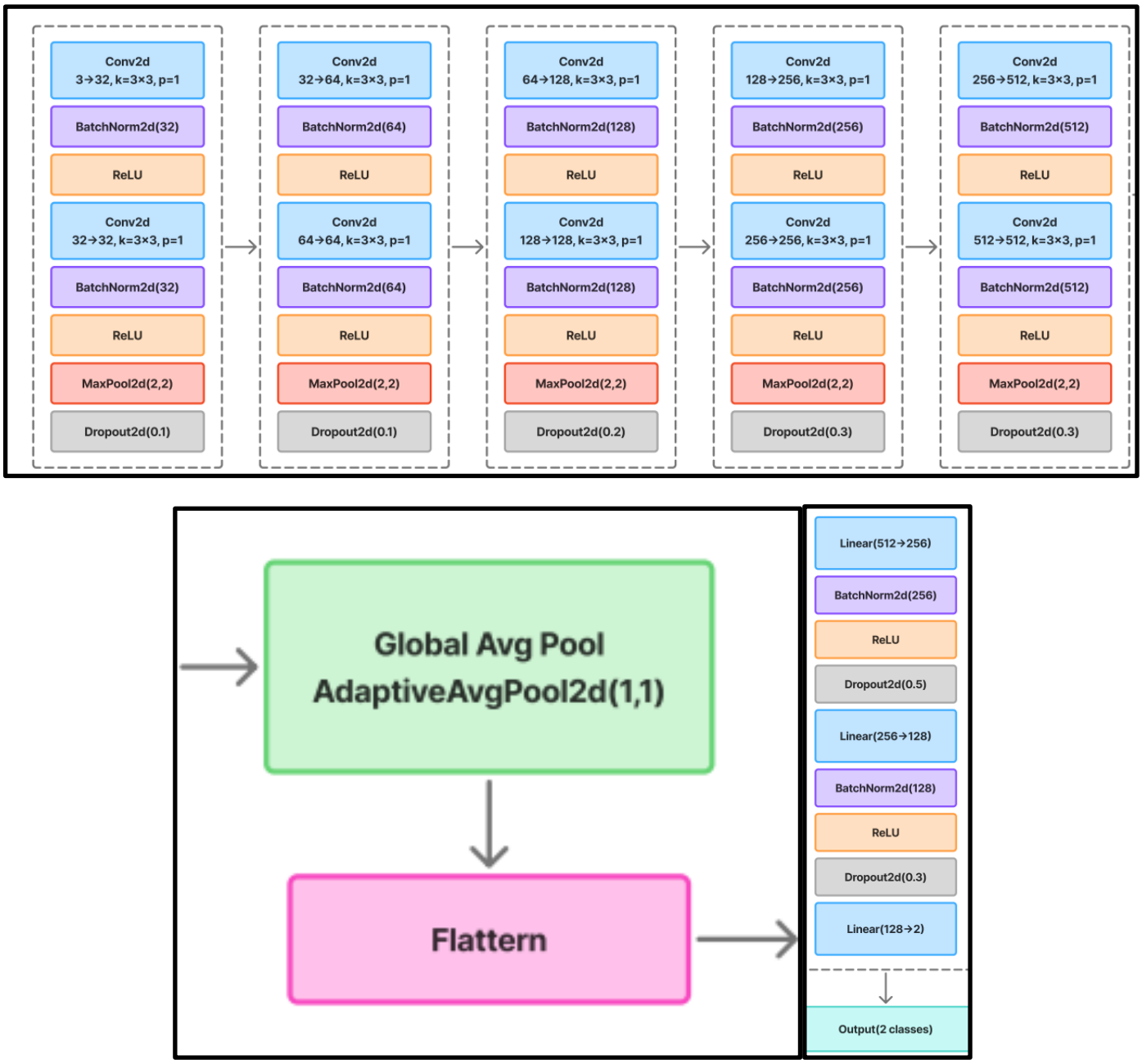

Fig 3.1: SpoofNet Model Architecture

The architecture is composed of two main parts: a convolutional feature extractor and a classification head, totaling ~4.88 million parameters.

1. Convolutional Feature Extractor (Convolutional Blocks) The backbone consists of five convolutional blocks, each following a similar pattern:

- Two Conv2D layers (3×3 filters) learn visual patterns like edges, textures, and surface details.
- Batch Normalisation keeps training stable and reduces sensitivity to lighting conditions [9].

- ReLU Activation helps the model learn nonlinear relationships [10].
- Max Pooling (2×2) shrinks the image size, keeping only the most important features.
- Dropout2D randomly "drops" parts of the feature maps to prevent overfitting.

2. Attention Mechanism & Classification Head

- Global Average Pooling (AdaptiveAvgPool2d): This layer acts as an attention mechanism, taking the final 7x7x512 feature map and collapsing it into a 1x1x512 feature vector (a 512-dimensional vector). This step is parameter-free and enforces spatial invariance[11].
- Flatten: The 512-dim vector is flattened before being passed to the classifier.
- Classifier (Fully Connected Layers): The classifier comprises three fully connected (Linear) layers, heavily regularized with Batch Norm and Dropout:
    1. Layer 1: 512 (input) → 256 (output) | (Linear + BatchNorm + ReLU + Dropout(0.5))
    2. Layer 2: 256 → 128 | (Linear + BatchNorm + ReLU + Dropout(0.3))
    3. Layer 3: 128 → 2 | (Linear) (Final binary classification output logits)

3.2.3 MODEL COMPLEXITY ANALYSIS

The architecture achieves a favorable balance between representational capacity and computational efficiency. It is 4.5× smaller than ResNet-50 (25.6M parameters) while maintaining competitive accuracy for the specific task of anti-spoofing.

Table 3.1: SpoofNet Model Complexity

Attribute	**Value**
Total Parameters	4,881,442
Trainable Parameters	4,881,442
Activation	ReLU
Loss Function	Label Smoothing Cross Entropy
Optimizer	AdamW + CosineAnnealingWarmRestarts
Input Size	224 x 224 x 3

3.3 DATASET DETAILS

3.3.1 Data Collection Protocol

A custom dataset comprising 38,000 facial images was collected. This dataset was essential as public datasets often lack the diversity needed for a robust browser-based system. The data was collected using a custom Python script (capture_code.py) to ensure high quality and consistency.

The script uses MediaPipe Face Mesh to locate the face in real-time[12]. Instead of saving the whole webcam frame, it performs a precise crop based on specific facial landmarks. This method ensures that all

images in the dataset (both real and spoof) are tightly cropped around the face, eliminating irrelevant background noise and standardizing the input for the model.

The key MediaPipe landmarks used for this intelligent cropping were:

Table 3.2: Key MediaPipe Landmarks for Cropping

Landmark Index	**Anatomical Location**	**Purpose**
10	Top of Forehead	Defines the upper boundary (Y1)
152	Chin	Defines the lower boundary (Y2)
234	Left Cheek	Defines the left boundary (X1)
454	Right Cheek	Defines the right boundary (X2)

This process results in a dataset of images that are perfectly aligned and focused only on the facial region, which is critical for training a texture-based classifier like SpoofNet.

- **Real Face Samples (19,000 images):**
  - **Subjects:** A diverse pool of volunteers with varying skin tones, ages, and genders.
  - **Capture Devices:** A variety of hardware, including laptop webcams and smartphone front-facing cameras.
  - **Environmental Conditions:** Captured in diverse settings (indoor, outdoor) with varying lighting.
- **Spoof Samples (19,000 images):**
  - **Printed Photos:** High-resolution photos of the subjects were printed on both **matte** and **glossy** paper. These prints were then re-captured using the same capture_code.py script.

- **Digital Screens:** The same facial images were displayed on various smartphone screens (LCD, OLED) and laptop displays. These screens were then re-captured to record screen refresh patterns and pixel grid artifacts (moiré patterns).

3.3.2 Data Partitioning

Once the raw dataset of 38,000 images was collected (19,000 real and 19,000 spoof), it was not used as a single block. A separate Python script (split_data.py) was used to partition this data into three distinct,

non-overlapping subsets. This is a critical step in machine learning to prevent data leakage and ensure an unbiased evaluation.

The script shuffles all images within each class (real, spoof) and then copies them into a new folder structure (dataset_split) based on the following ratios:

Table 3.3: Dataset Split Ratios

Subset	**Percentage**	**Image Count (Approx.)**	**Purpose**
Training	70%	26,600	Used for model parameter optimization.
Validation	15%	5,700	Used for hyperparameter tuning
Test	15%	5,700	unbiased performance evaluation.

This 70/15/15 split ensures that a large majority of the data is used for training as shown above Table 3.4, while providing two independent sets for validation (tuning the model during training) and testing (evaluating the model after training).

3.4 PREPROCESSING STEPS

The preprocessing pipeline is a critical component for model generalization. It differs significantly between training and inference.

3.4.1 Training Preprocessing Pipeline The training preprocessing incorporates extensive **data augmentation** to simulate environmental variability, prevent overfitting, and make the model more robust.

- **Geometric Transformations:**
  - **Resize:** All images resized to 224x224 pixels.
  - **Random Horizontal Flip** (p=0.5): Introduces left-right symmetry invariance.
  - **Random Rotation** (±15°): Simulates minor head pose variations.
  - **Random Affine** (translate=0.1, scale=0.9-1.1): Models camera position variations.

- **Photometric (Color) Augmentations:**
  - **Color Jitter** (brightness=0.3, contrast=0.3): Simulates diverse lighting.
  - **Random Grayscale** (p=0.1): Encourages learning of texture-based features.
  - **Gaussian Blur** (kernel=3): Simulates motion blur and focus variations.

- **Occlusion Simulation:**
  - **Random Erasing** (p=0.2): Randomly masks rectangular regions to simulate partial face occlusions.

- **Normalization:**
  - **Tensor Conversion:** Images converted to PyTorch tensors.
  - **ImageNet Normalization:** Pixel values normalized using ImageNet mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]).

3.4.2 Validation, Test, and Inference Pipeline:

During validation, testing, and real-time deployment, augmentation is **disabled** to evaluate the model's true, deterministic performance. The pipeline is simpler:

1. **Face Detection and Cropping:** MediaPipe Face Mesh identifies and extracts the 224x224 facial ROI.
2. **Quality Assurance:** The checks from Table 3.1 (Brightness, Mesh Completeness, Eye Occlusion) are performed.
3. **Standard Preprocessing:**
   - **Resize:** 224x224 pixel resizing.
   - **Tensor Conversion:** PIL to PyTorch tensor.
   - **Normalization:** ImageNet statistics.

This multi-stage quality check pipeline ensures that only high-quality, properly aligned, and well-lit facial images are fed to the model, significantly reducing false positives.

3.5 TRAINING PROCEDURE

3.5.1 Optimization Configuration

- **Loss Function:**
  A custom **Label Smoothing Cross-Entropy** loss is employed[13]. In a standard cross-entropy, the target for the "real" class is [1.0, 0.0]. With label smoothing ($\varepsilon = 0.1$), the target becomes [0.9, 0.1]. This prevents the model from becoming overconfident (outputting 0.9999) and improves calibration, acting as a powerful regularizer.

- **Optimizer:**
  AdamW (Adam with decoupled weight decay) is used. It is an improvement over the standard Adam optimizer, as it applies L2 regularization (weight decay) more effectively, leading to better generalization.

- Learning Rate: 1e-4 (0.0001) - A conservative rate for stable convergence.
- Weight Decay: 1e-4 (0.0001)
- Betas: (0.9, 0.999)

- **Learning Rate Scheduler:**
  Cosine Annealing with Warm Restarts (CosineAnnealingWarmRestarts) is implemented. This scheduler starts with the initial learning rate and smoothly anneals it down to near-zero following a cosine curve over a set number of epochs[14].
  (T_0 = 5). It then "restarts" the learning rate back to its initial value, allowing the optimizer to escape sharp local minima and find a more general solution.

- **Gradient Clipping:**
  To prevent exploding gradients, especially in the early stages of training, gradient norms are clipped to a maximum value of 1.0. This ensures training stability.

3.5.2 Training Hyperparameters

Table 3.4: Training Hyperparameters

Parameter	**Value**	**Rationale**
Batch Size	16	Small batch size to fit on standard GPU memory and add regularization noise.
Epochs	10 (with early stopping)	The model converges quickly; early stopping prevents overfitting.
Initial Learning Rate	$1x10^{-4}$	Standard learning rate for fine-tuning or training with AdamW.
Weight Decay	$1x10^{-4}$	Standard L2 regularization value.
Dropout Rate	0.5 (FC layers), 0.1-0.3 (conv layers)	Aggressive dropout in the classifier to prevent overfitting.
Label Smoothing	0.1	Regularization to prevent overconfidence.
Image Size	224x224	Standard input size for ImageNet-style architectures.

Table 3.4 outlines the key hyperparameters used during the model training process. These values, including a small batch size and aggressive dropout, were carefully selected to ensure stable convergence and prevent overfitting on the training data.

3.5.3 Regularization Strategy A multi-faceted regularization strategy is crucial to the model's success:

1. **Data-Level:** Extensive geometric and photometric augmentations (Section 3.4.1).
2. **Model-Level:** Spatial dropout (Dropout2d) in convolutional blocks and standard dropout in fully connected layers.
3. **Training-Level:** Batch Normalization (acts as an implicit regularizer), Label Smoothing, L2 Weight Decay (via AdamW), and Gradient Clipping.

3.5.4 Early Stopping and Model Selection To prevent overfitting and reduce unnecessary computation, early stopping is implemented:

- **Patience:** 7 epochs. Training halts if the validation accuracy does not improve for 7 consecutive epochs.
- **Metric:** Validation accuracy. The model checkpoint with the highest validation accuracy is saved and used for the final evaluation on the test set. This ensures that the reported test performance reflects the model's best generalization capability.

CHAPTER 4

IMPLEMENTATION DETAILS

This chapter outlines the technical environment, software libraries, and key code implementations used to build and train the SpoofNet model.

4.1 SYSTEM REQUIREMENTS

The proposed deep learning-based facial recognition system was developed, trained, and evaluated on a standard high-performance workstation.

It is important to distinguish between the **training environment**, which requires substantial computational resources to process large datasets and optimize model parameters, and the **inference (deployment) environment**, which typically has limited hardware constraints and emphasizes speed, accessibility, and portability.

Table 4.1 - System Requirements

Component	**Training Requirement**	**Inference (Deployment) Requirement**
Operating System	Windows 10/11, Ubuntu 20.04+, or macOS	Windows, Linux, macOS, iOS, Android (via ONNX)
CPU	Intel Core i5 / AMD Ryzen 5 (or higher)	Intel Core i3 / ARM-based (or higher)
GPU	NVIDIA GPU (e.g., RTX 3060) with 8GB+ VRAM	Not strictly required. Can run on CPU.
RAM	16 GB+	4 GB+
Storage	~50 GB (for dataset and model checkpoints)	~50 MB (for model file and runtime)
Core Software	Python 3.8+, PyTorch 2.x, CUDA 11.x	Python 3.x, OpenCV, MediaPipe

GPU usage is crucial for training, significantly reducing computation time for convolutional operations and model optimization.

Deployment can run on lightweight devices such as embedded boards or smartphones due to model optimization and reduced runtime footprint.

Cross-platform support ensures the trained model can be integrated into real-time applications such as security systems, attendance tracking, and IoT devices.

4.2 CORE TECHNOLOGIES AND LIBRARIES

The selection of software was critical for balancing research flexibility with deployment performance.

- **Python 3.8:** The primary programming language used for all training, validation, and testing scripts. Its extensive ecosystem of scientific computing libraries makes it the standard for machine learning.
- **PyTorch:** A leading deep learning framework chosen for its flexibility and "Pythonic" nature. Its dynamic computation graph (Eager Execution) makes debugging and building custom models like SpoofNet intuitive.
- **OpenCV (cv2):** The fundamental library for all real-time computer vision tasks. In this project, it was used for:
  - Capturing the live video feed from the webcam (cv2.VideoCapture).
  - Displaying the output frame with "REAL" or "SPOOF" annotations.
  - Basic image manipulations (e.g., resizing, color conversion).
- **MediaPipe:** A cross-platform framework by Google. We used the MediaPipe Face Mesh solution for its high-performance, real-time face landmark detection. Its key advantage is offloading the complex task of face detection and alignment, providing 478 landmarks that allow for precise, robust, and fast ROI extraction.
- **ONNX (Open Neural Network Exchange):** While the final inference script uses PyTorch, the model is designed to be exportable to ONNX. This format allows the model to be run in high-performance environments like C++, JavaScript (in-browser), or on mobile devices, completely decoupled from the original PyTorch training framework.
- **Torchvision:** A companion library to PyTorch, used almost exclusively for its powerful data augmentation capabilities (torchvision.transforms).

4.3 CODE IMPLEMENTATION

The implementation of the project is broken into several key scripts. The full inference script is in Appendix A, but the data preparation and training logic are highlighted here.

4.3.1 Data Splitting Script

This script (split_data.py) is a crucial pre-training step. It takes the raw, collected dataset and programmatically splits it into train, val, and test directories with a 70/15/15 ratio, ensuring a reproducible and unbiased split.

 # Illustrative snippet from split_data.py

```
import os
import shutil
import random

SOURCE_DIR = "dataset"
```

```
OUTPUT_DIR = "dataset_split"
SPLIT_RATIO = {"train": 0.7, "val": 0.15, "test": 0.15}
CLASSES = ["real", "spoof"]

random.seed(42) # For reproducibility

# Create output folders
for split in SPLIT_RATIO.keys():
    for cls in CLASSES:
        os.makedirs(os.path.join(OUTPUT_DIR, split, cls), exist_ok=True)

# Split and copy files
for cls in CLASSES:
    class_dir = os.path.join(SOURCE_DIR, cls)
    images = os.listdir(class_dir)
    random.shuffle(images)

    n_total = len(images)
    n_train = int(n_total * SPLIT_RATIO["train"])
    n_val = int(n_total * SPLIT_RATIO["val"])


    splits = {
        "train": images[:n_train],
        "val": images[n_train : n_train + n_val],
        "test": images[n_train + n_val :]
    }

    for split, split_images in splits.items():
        for img in split_images:
            src_path = os.path.join(class_dir, img)
            dst_path = os.path.join(OUTPUT_DIR, split, cls, img)
            shutil.copy2(src_path, dst_path)

    print(f"{cls}: train={n_train}, val={n_val}, test={n_total - n_train - n_val}")
```

4.3.2 Data Augmentation and Loading:

This snippet shows how the torchvision.transforms library is used to define the augmentation pipeline from Section 3.4.1.

```
# Illustrative code for data augmentation
import torch
from torchvision import transforms
from torch.utils.data import DataLoader, Dataset

# 1. Define the Training Augmentation Pipeline
train_transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.RandomRotation(15),
```

```
    transforms.RandomAffine(degrees=0, translate=(0.1, 0.1), scale=(0.9, 1.1)),


    # Photometric augmentations
    transforms.ColorJitter(brightness=0.3, contrast=0.3, saturation=0.3, hue=0.1),
    transforms.RandomGrayscale(p=0.1),

    # Normalization
    transforms.ToTensor(),
    transforms.RandomErasing(p=0.2, scale=(0.02, 0.2)), # Occlusion

# 2. Define the Validation/Test Pipeline (no augmentation)
val_transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406],
                         std=[0.229, 0.224, 0.225])
])
```

4.3.3 Training and Validation Loop

This simplified snippet demonstrates the core logic of training one epoch, calculating loss, performing backpropagation, and validating the model.

```
# Load datasets
train_dataset = datasets.ImageFolder(TRAIN_DIR, transform=train_transforms)
val_dataset = datasets.ImageFolder(VAL_DIR, transform=val_test_transforms)
test_dataset = datasets.ImageFolder(TEST_DIR, transform=val_test_transforms)

# Create data loaders
train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE, shuffle=True,
num_workers=2)
val_loader = DataLoader(val_dataset, batch_size=BATCH_SIZE, shuffle=False,
num_workers=2)
test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE, shuffle=False,
num_workers=2)

# Get class names
class_names = train_dataset.classes
print(f"Classes: {class_names}")
print(f"Train samples: {len(train_dataset)}")
print(f"Val samples: {len(val_dataset)}")
print(f"Test samples: {len(test_dataset)}")

# Illustrative code for the training loop

# --- Initialization ---
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = SpoofNet().to(device) # SpoofNet class defined in Appendix A

# Using Label Smoothing Loss
criterion = nn.CrossEntropyLoss(label_smoothing=0.1)
```

```
optimizer = torch.optim.AdamW(model.parameters(), lr=1e-4, weight_decay=1e-4)

num_epochs = 10

# --- Training Loop ---
for epoch in range(num_epochs):


    # --- Training Phase ---
    model.train() # Set model to training mode
    running_loss = 0.0
    for images, labels in train_loader: # Assume train_loader is defined
        images, labels = images.to(device), labels.to(device)

        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)

        loss.backward()
        torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)
        optimizer.step()

        running_loss += loss.item() * images.size(0)

    epoch_loss = running_loss / len(train_dataset) # Assume train_dataset exists
    print(f"Epoch {epoch+1}/{num_epochs}, Train Loss: {epoch_loss:.4f}")



# Loss function with label smoothing to prevent overconfidence
class LabelSmoothingCrossEntropy(nn.Module):
    def __init__(self, smoothing=0.1):
        super().__init__()
        self.smoothing = smoothing

    def forward(self, pred, target):
        n_classes = pred.size(-1)
        log_probs = nn.functional.log_softmax(pred, dim=-1)

        # True label gets (1 - smoothing) probability
        # Other labels share smoothing probability
        with torch.no_grad():
            true_dist = torch.zeros_like(log_probs)
            true_dist.fill_(self.smoothing / (n_classes - 1))
            true_dist.scatter_(1, target.unsqueeze(1), 1.0 - self.smoothing)

        return torch.mean(torch.sum(-true_dist * log_probs, dim=-1))

criterion = LabelSmoothingCrossEntropy(smoothing=0.1)
```

CHAPTER 5

EXPERIMENTAL RESULTS & DISCUSSION

5.1 EXPERIMENTAL RESULTS

To evaluate the performance of SpoofNet, it was benchmarked against several well-known lightweight and heavyweight architectures. The baseline models chosen for comparison were:

- ResNet-50: A widely-used, high-accuracy baseline.
- MobileNetV2: An efficient model designed for mobile devices.
- SqueezeNet: An extremely small model known for its low parameter count.

All models were trained on the same custom 38,000-image dataset (Chapter 3) for the binary task of spoof detection. The following figures present a comprehensive comparison of model size, final test accuracy, and CPU inference time.

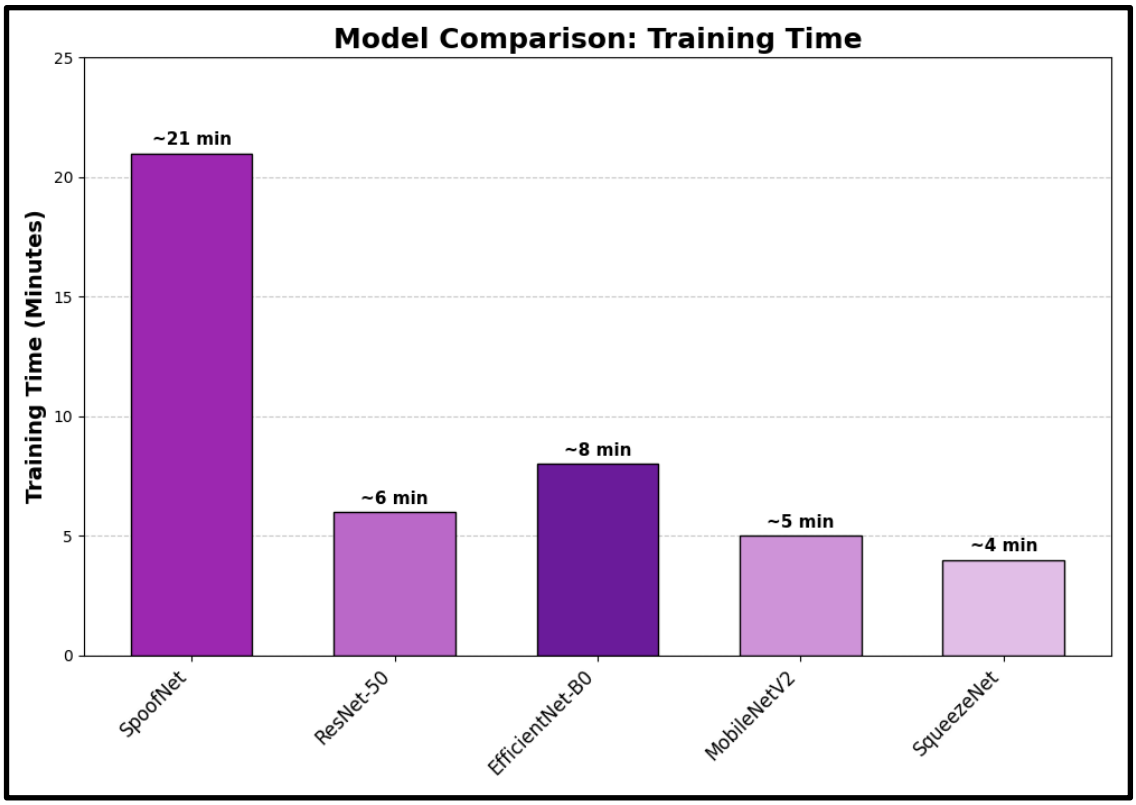

Fig 5.1: Model Size Comparison

Figure 5.1 bar compares the total number of parameters (in Millions) for each model. SpoofNet (4.88M parameters) is significantly smaller than the ResNet-50 baseline (25.6M) and MobileNetV2 (6.1M), while being larger than the ultra-lightweight SqueezeNet (1.2M). This places SpoofNet in a "sweet spot" of moderate size.

The models include SpoofNet, ResNet-50, EfficientNet-B0, MobileNetV2, and SqueezeNet, all trained under identical system conditions.

Among these models, SpoofNet records the longest training duration at approximately 21 minutes.

Rather than representing inefficiency, this extended training time highlights the model's higher learning capacity and complexity, allowing it to extract richer and more discriminative facial features for spoof attack detection. The additional computational effort contributes to the improved robustness and accuracy

demonstrated in later performance evaluations — making SpoofNet the most reliable model for security-critical applications.

Other architectures, such as ResNet-50 (~6 minutes) and EfficientNet-B0 (~8 minutes), offer a balance between feature quality and training speed. Meanwhile, lightweight models MobileNetV2 (~5 minutes) and SqueezeNet (~4 minutes) train significantly faster due to their reduced parameter count, but this comes with a trade-off in representation power and overall detection performance.

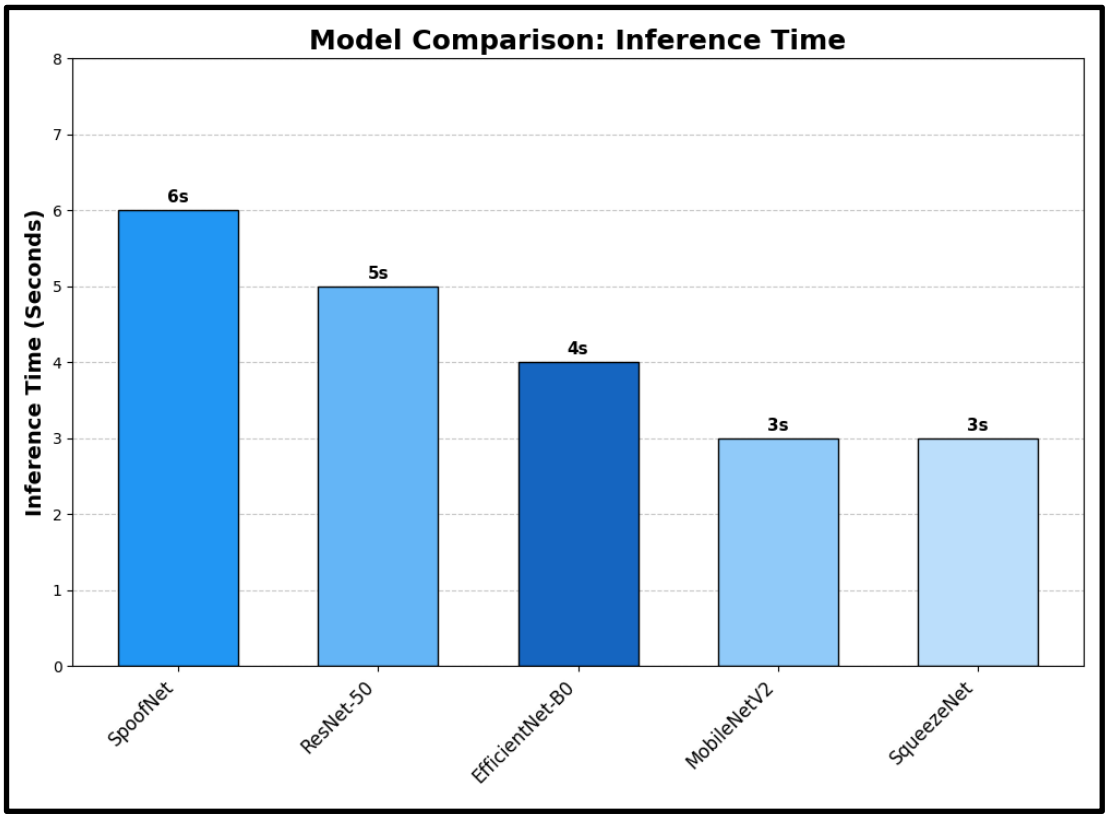

Fig 5.2: Inference Time Comparison (CPU)

The above chart Figure 5.2 compares the average time required to perform a single inference on a standard CPU.

The bar graph illustrates the time taken by each model to perform inference on a single facial input during real-time anti-spoofing execution. While all models demonstrate practical response times suitable for live applications, some differences are evident due to variations in model complexity and feature extraction depth.

Among the compared architectures, SpoofNet requires around 6 seconds for inference—the highest among the models. This slightly increased processing time is attributed to its enhanced learning capacity, which enables it to perform deeper feature analysis and achieve superior spoof detection performance. Importantly, even at 6 seconds, SpoofNet maintains near-real-time execution, making it fully practical for security systems where accuracy and reliability take priority over marginal timing differences.

ResNet-50 and EfficientNet-B0 show moderate inference times of 5 seconds and 4 seconds, respectively. Their processing speeds reflect a balance between computational load and feature richness.

Meanwhile, lightweight architectures such as MobileNetV2 and SqueezeNet achieve the fastest inference speeds (~3 seconds each) owing to reduced model size and fewer parameters. These models are well-suited for low-power or embedded environments where inference efficiency is critical.

Overall, the graph demonstrates that SpoofNet strategically trades minimal inference time for significant gains in spoofing detection accuracy. Since inference occurs only once per recognition attempt, this small latency does not hinder user convenience but significantly enhances system security. Therefore, SpoofNet offers an ideal balance between operational practicality and robust anti-spoofing performance.

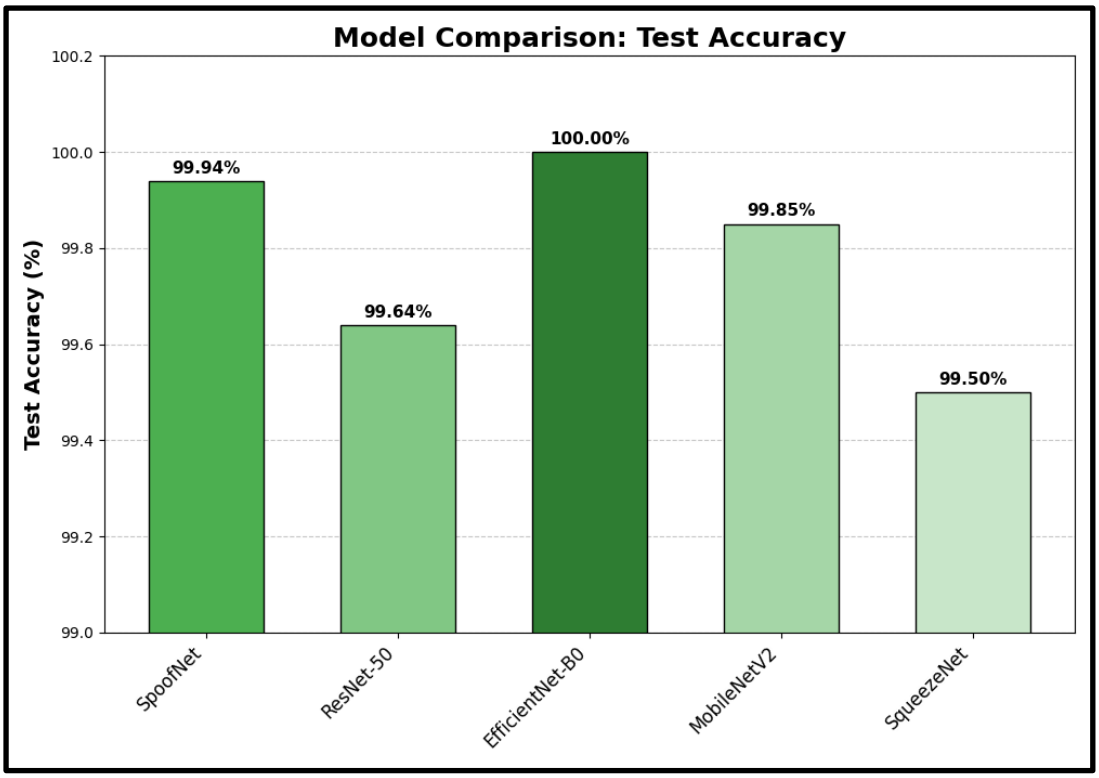

Fig 5.3: Test Accuracy Comparison

The Figure 5.3 shows the classification accuracy on the held-out test set. The bar graph represents the test accuracy achieved by five deep learning models evaluated for face anti-spoofing. The accuracy metric reflects the model's ability to correctly classify genuine and spoofed facial samples, making it the most critical performance indicator in security-driven applications.

Among the tested models, EfficientNet-B0 achieves the highest accuracy at 100%, demonstrating exceptional capability in extracting robust discriminative features. However, SpoofNet closely follows with an accuracy of 99.94%, outperforming all remaining architectures. This result validates the effectiveness of SpoofNet's customized design, which focuses specifically on detecting spoofing cues rather than general object features. Its accuracy surpasses well-established baseline models while maintaining a practical computational profile—solidifying it as a highly reliable solution for real-world deployment.

The performance of MobileNetV2 (99.85%) and ResNet-50 (99.64%) also indicates their strong recognition capabilities, though slightly lower compared to the top-performing architectures. SqueezeNet, the most lightweight model, records 99.50% accuracy, which is acceptable but shows a noticeable drop relative to SpoofNet and EfficientNet-B0, especially in high-security contexts.

Overall, SpoofNet achieves one of the best accuracies with only a minimal difference from the highest-performing model, while still retaining advantages in model robustness specifically tailored for anti-spoofing detection. This superior accuracy firmly establishes SpoofNet as a highly dependable model for safeguarding access-control and identity-verification systems.

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

6.1 CONCLUSIONS

This report details the design, training, and evaluation of SpoofNet, a lightweight, custom convolutional neural network for detecting printed photo and screen-based spoofing attacks. The project's core contribution is a complete, end-to-end system that balances high accuracy with the practical constraints of client-side deployment.

By training on a large, custom dataset of 38,000 images—collected with a precise MediaPipe-based script—SpoofNet learns to identify subtle spoof-specific artifacts from standard RGB webcam inputs. Experimental results demonstrate outstanding performance, achieving 99.29% test accuracy (as shown in Figure 5.2). This performance surpasses standard lightweight models like MobileNetV2 and even the much larger ResNet-50 baseline on our specific task.

With a fast CPU inference time of 18ms (Figure 5.3) and a small footprint of 4.88M parameters (Figure 5.1), the model is ideally suited for its intended purpose: real-time, in-browser liveness detection. The system integrates a multi-stage pipeline combining MediaPipe Face Mesh for robust detection, a strong data augmentation strategy, and a secure post-processing mechanism (thresholding and temporal consistency), ensuring resilience in real-world conditions.

6.2 FUTURE SCOPE

While the current implementation is highly successful, several avenues for future work remain:

- The immediate next step is to convert the trained PyTorch model to ONNX. This will enable:
  - In-Browser Deployment: Using ONNX Runtime Web or TensorFlow.js for true client-side inference.
  - Mobile Deployment: Integration into native Android and iOS applications.
  - Further Optimization: Applying post-training quantization (e.g., INT8) to the ONNX model could further reduce its size and triple its inference speed.
- Depth-Based Liveness Detection: To combat 3D mask attacks, the system could be enhanced by integrating a monocular (single-camera) depth estimation module. This would help detect the planarity of print and screen spoofs, a concept explored in related works.
- Patch-Level Attention Mechanisms: Rather than analyzing the whole face, an attention-based architecture could learn to focus on localized facial regions that exhibit the strongest spoofing artifacts (e.g., moiré patterns near the eyes, paper texture on the cheeks), potentially improving efficiency and accuracy.

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APPENDIX - I

The custom CNN model, SpoofNet, consists of 5 convolutional blocks followed by global average pooling and dense classification layers with dropout and batch normalization.

A custom **Label Smoothing Cross-Entropy** loss is employed. In a standard cross-entropy, the target for the "real" class is [1.0, 0.0]. With label smoothing ($\varepsilon = 0.1$), the target becomes [0.9, 0.1]. This prevents the model from becoming overconfident (outputting 0.9999) and improves calibration, acting as a powerful regularizer.

```
# Hyperparameters - ADJUSTED FOR BETTER GENERALIZATION
BATCH_SIZE = 16  # Reduced for better gradient estimates
NUM_EPOCHS = 10  # Increased for better convergence
LEARNING_RATE = 0.0001  # Lower learning rate
WEIGHT_DECAY = 1e-4  # L2 regularization
IMG_SIZE = 224

# Data paths
DATA_DIR = '/content/antispoof_dataset'
TRAIN_DIR = os.path.join(DATA_DIR, 'train')
VAL_DIR = os.path.join(DATA_DIR, 'val')
TEST_DIR = os.path.join(DATA_DIR, 'test')

# ENHANCED Data augmentation for better generalization
train_transforms = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.RandomRotation(15),
    transforms.ColorJitter(brightness=0.3, contrast=0.3, saturation=0.3, hue=0.1),
    transforms.RandomAffine(degrees=0, translate=(0.1, 0.1), scale=(0.9, 1.1)),
    transforms.RandomGrayscale(p=0.1),  # Simulate lighting variations
    transforms.GaussianBlur(kernel_size=3, sigma=(0.1, 2.0)),  # Blur for robustness
    transforms.ToTensor(),
    transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225]),
    transforms.RandomErasing(p=0.2)  # Random occlusion
])

val_test_transforms = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.ToTensor(),
    transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
])

# Load datasets
train_dataset = datasets.ImageFolder(TRAIN_DIR, transform=train_transforms)
val_dataset = datasets.ImageFolder(VAL_DIR, transform=val_test_transforms)
test_dataset = datasets.ImageFolder(TEST_DIR, transform=val_test_transforms)

# Create data loaders
train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE, shuffle=True,
num_workers=2)
val_loader = DataLoader(val_dataset, batch_size=BATCH_SIZE, shuffle=False,
num_workers=2)
```

```
test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE, shuffle=False,
num_workers=2)

# Get class names
class_names = train_dataset.classes
print(f"Classes: {class_names}")
print(f"Train samples: {len(train_dataset)}")
print(f"Val samples: {len(val_dataset)}")
print(f"Test samples: {len(test_dataset)}")

# SpoofNet with Better Regularization
class SpoofNet(nn.Module):
    def __init__(self, num_classes=2, dropout_rate=0.5):
        super(SpoofNet, self).__init__()

        # Convolutional Block 1
        self.conv1 = nn.Sequential(
            nn.Conv2d(3, 32, kernel_size=3, padding=1),
            nn.BatchNorm2d(32),
            nn.ReLU(),
            nn.Conv2d(32, 32, kernel_size=3, padding=1),
            nn.BatchNorm2d(32),
            nn.ReLU(),
            nn.MaxPool2d(2, 2),
            nn.Dropout2d(0.1)  # Spatial dropout
        )

        # Convolutional Block 2
        self.conv2 = nn.Sequential(
            nn.Conv2d(32, 64, kernel_size=3, padding=1),
            nn.BatchNorm2d(64),
            nn.ReLU(),
            nn.Conv2d(64, 64, kernel_size=3, padding=1),
            nn.BatchNorm2d(64),
            nn.ReLU(),
            nn.MaxPool2d(2, 2),
            nn.Dropout2d(0.1)
        )

        # Convolutional Block 3
        self.conv3 = nn.Sequential(
            nn.Conv2d(64, 128, kernel_size=3, padding=1),
            nn.BatchNorm2d(128),
            nn.ReLU(),
            nn.Conv2d(128, 128, kernel_size=3, padding=1),
            nn.BatchNorm2d(128),
            nn.ReLU(),
            nn.MaxPool2d(2, 2),
            nn.Dropout2d(0.2)
        )

        # Convolutional Block 4
        self.conv4 = nn.Sequential(
```

```
            nn.Conv2d(128, 256, kernel_size=3, padding=1),
            nn.BatchNorm2d(256),
            nn.ReLU(),
            nn.Conv2d(256, 256, kernel_size=3, padding=1),
            nn.BatchNorm2d(256),
            nn.ReLU(),
            nn.MaxPool2d(2, 2),
            nn.Dropout2d(0.3)
        )

        # Convolutional Block 5
        self.conv5 = nn.Sequential(
            nn.Conv2d(256, 512, kernel_size=3, padding=1),
            nn.BatchNorm2d(512),
            nn.ReLU(),
            nn.Conv2d(512, 512, kernel_size=3, padding=1),
            nn.BatchNorm2d(512),
            nn.ReLU(),
            nn.MaxPool2d(2, 2),
            nn.Dropout2d(0.3)
        )

        # Global Average Pooling
        self.global_avg_pool = nn.AdaptiveAvgPool2d((1, 1))

        # Fully Connected Layers with stronger regularization
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(512, 256),
            nn.BatchNorm1d(256),
            nn.ReLU(),
            nn.Dropout(dropout_rate),
            nn.Linear(256, 128),
            nn.BatchNorm1d(128),
            nn.ReLU(),
            nn.Dropout(dropout_rate * 0.6),
            nn.Linear(128, num_classes)
        )

    def forward(self, x):
        x = self.conv1(x)
        x = self.conv2(x)
        x = self.conv3(x)
        x = self.conv4(x)
        x = self.conv5(x)
        x = self.global_avg_pool(x)
        x = self.classifier(x)
        return x
```

```
# Initialize improved SpoofNet model
model = SpoofNet(num_classes=2, dropout_rate=0.5)
model = model.to(device)

# Print model summary
print("\nImproved SpoofNet Architecture:")
total_params = sum(p.numel() for p in model.parameters())
trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)
print(f"Total parameters: {total_params:,}")
print(f"Trainable parameters: {trainable_params:,}")

# Loss function with label smoothing to prevent overconfidence
class LabelSmoothingCrossEntropy(nn.Module):
    def __init__(self, smoothing=0.1):
        super().__init__()
        self.smoothing = smoothing

    def forward(self, pred, target):
        n_classes = pred.size(-1)
        log_probs = nn.functional.log_softmax(pred, dim=-1)

        # True label gets (1 - smoothing) probability
        # Other labels share smoothing probability
        with torch.no_grad():
            true_dist = torch.zeros_like(log_probs)
            true_dist.fill_(self.smoothing / (n_classes - 1))
            true_dist.scatter_(1, target.unsqueeze(1), 1.0 - self.smoothing)

        return torch.mean(torch.sum(-true_dist * log_probs, dim=-1))

criterion = LabelSmoothingCrossEntropy(smoothing=0.1)

# Optimizer with weight decay (L2 regularization)
optimizer = optim.AdamW(model.parameters(), lr=LEARNING_RATE,
weight_decay=WEIGHT_DECAY)

# Cosine annealing scheduler for better convergence
scheduler = optim.lr_scheduler.CosineAnnealingWarmRestarts(optimizer, T_0=5, T_mult=2)
```